

Stacks

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Outline

- 1 Definition
- 2 Implementation



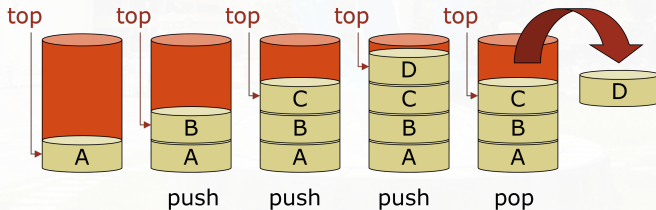
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Definition

- A stack is an ordered list in which **insertions** and deletions are made at the end “top”.
 - insertions: push/add
 - deletions: pop/remove
- Last-In-First-Out (LIFO).



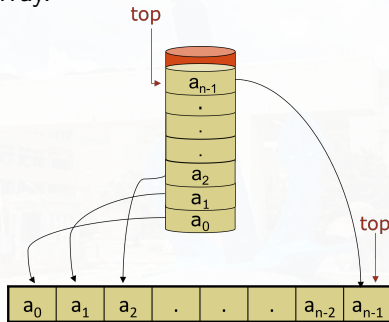
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Stack Implementation: Array

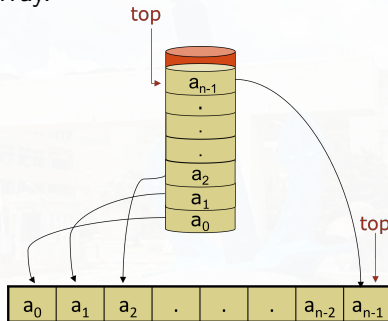
- The easiest way to implement the stack ADT is using one-dimensional array.



- An example in C++

Stack Implementation: Array

- The easiest way to implement the stack ADT is using one-dimensional array.



- An example in C++ (another way: using a linked list; will be introduced in the future).

Functions for Stacks

- Create a stack.
 - Create an empty stack with maximum size MAX_STACK_SIZE.

```
#define MAX_STACK_SIZE 101

typedef struct {
    int key; // can be of other types...
    /* other fields? */
} element;

element stack[MAX_STACK_SIZE];
int top = -1; // initially no element
```



Functions for Stacks (2/2)

- **IsEmpty**
 - Return true if the stack is empty and false otherwise.



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 $top < 0$
- **IsFull**
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Functions for Stacks (2/2)

- IsEmpty

- Return true if the stack is empty and false otherwise.

$top < 0$

- IsFull

- Return true if the stack is full and false otherwise.

$top \geq MAX_STACK_SIZE-1;$

- Push (or Add)

- Insert the element into the top of the stack.



Functions for Stacks (2/2)

- IsEmpty

- Return true if the stack is empty and false otherwise.
`top < 0`

- IsFull

- Return true if the stack is full and false otherwise.
`top >= MAX_STACK_SIZE-1;`

- Push (or Add)

- Insert the element into the **top** of the stack.
`stack[++top] = element;`

- Pop (or Delete)

- Remove and return the item on the top of the stack.



Functions for Stacks (2/2)

- IsEmpty

- Return true if the stack is empty and false otherwise.
`top < 0`

- IsFull

- Return true if the stack is full and false otherwise.
`top >= MAX_STACK_SIZE-1;`

- Push (or Add)

- Insert the element into the `top` of the stack.
`stack[++top] = element;`

- Pop (or Delete)

- Remove and return the item on the top of the stack.
`return stack[top--];`



isFull & isEmpty

```
bool isEmpty() {  
    // note: bool type requires stdbool.h  
    if (top == -1) return true;  
    else return false;  
}  
  
bool isFull() {  
    if (top >= MAX_STACK_SIZE-1) return true;  
    else return false;  
}
```



push & stackFull

```
void push(element item) {  
    // add an item to the global stack  
    if (top >= MAX_STACK_SIZE - 1) {  
        stackFull();  
    }  
    stack[++top] = item;  
}  
  
void stackFull() {  
    fprintf(stderr, "Stack is FULL!!");  
    exit(EXIT_FAILURE);  
}
```



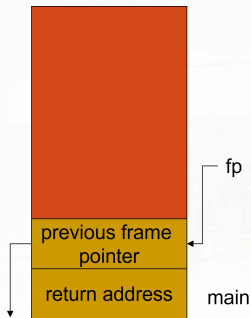
pop & stackEmpty

```
element pop() { // no argument is required!  
// return the top element from the stack  
    if (top == - 1) {  
        return stackEmpty(); // return an error key  
    }  
    return stack[top--];  
}  
  
element stackEmpty() {  
    element errKey; // depending the struct of element  
    errKey.key = -99;  
    fprintf(stderr, "Stack is EMPTY!!");  
    return errKey;  
}
```

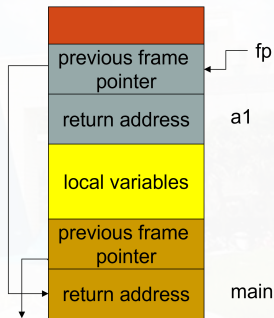


Supplementary: System Stack

- Stack frame of a function call



system stack **before** a_1 is invoked



system stack **after** a_1 is invoked



Easy Example: Summation from 1 to n

```
int sum(int n) {  
    element result;  
    result.key = 0;  
    element tmp;  
    tmp.key = n;  
    push(tmp); // bootstrapping here!  
  
    while (!isEmpty()) {  
        tmp = pop(stack);  
        result.key += tmp.key;  
        tmp.key = tmp.key-1;  
        if (tmp.key > 0) push(tmp);  
    }  
    return result.key;  
}
```



Left Home Exercise: Fibonacci

- Try to use a stack to turn the following recursive function into a non-recursive one.

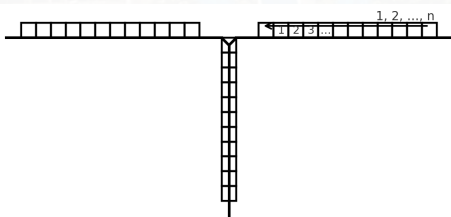
```
int Fibo(int n) {  
    if (n == 0) return 0;  
    else if (n == 1) return 1;  
    else return Fibo(n-1) + Fibo(n-2);  
}
```

- Another good exercise: implement the Ackermann function using a stack.



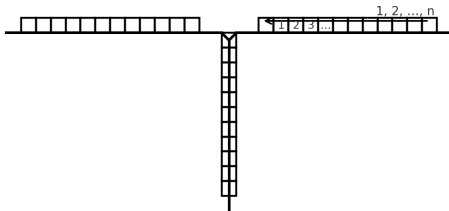
Exercise: Railroad Switching Network (1/2)

- Consider the railroad switching network given in the below figure. Railroad cars numbered $1, 2, 3, \dots, n$ are initially in the top right track segment.
- Railroad cars can be moved into the **vertical track segment** one at a time from either of the horizontal segments and then moved from the vertical segment to any one of the horizontal segments.
- The vertical segment operates as a **stack** as new cars enter at the top and cars depart the vertical segment from the top.



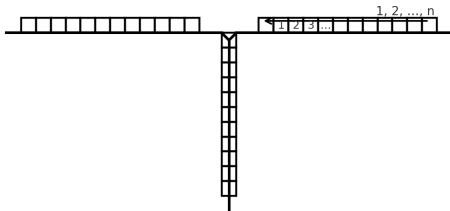
Exercise: Railroad Switching Network (2/2)

- For $n = 3$ and $n = 4$, what are the possible permutations of the cars that can be obtained? Are any permutations not possible?



Exercise: Railroad Switching Network (2/2)

- For $n = 3$ and $n = 4$, what are the possible permutations of the cars that can be obtained? Are any permutations not possible?



- What if the stack has the capacity limit (say, two)?

Discussions

